KPPSC PMS Geography Paper-I — Q.4: Weathering Processes (Physical & Chemical) & Mass Wasting

KPPSC PMS Master Examination Notes — Paper-I (Physical & General Geography)

Target Examination	KPPSC PMS (Provincial Management Service) — Khyber Pakhtunkhwa
Paper & Subject	Geography Paper-I (Physical & General Geography — 100 Marks)
EXAM QUESTIONS LINKED:	• **2022 Q.1:** "Discuss various processes of physical and chemical weathering with support from sketches/diagrams. Explain their geomorphic significance. (20 Marks)" • **2018 Q.1:** "Discuss the endogenic and exogenic geomorphic processes in detail. How do they shape the Earth's relief? (20 Marks)" • **2013 Q.4:** "Discuss the factors promoting mass movements in weathered materials. Explain the various types of mass movements. (20 Marks)"
Target Syllabus	Section 2: Landforms — Weathering (Physical & Chemical) & Mass Wasting
Marks & Allocation	20 Marks | Strict 35-Minute Descriptive Writing Limit
Core Theoretical Thesis	Weathering operates as the in-situ static disintegration (mechanical/physical) and decomposition (chemical) of bedrock, preparing material for geomorphic mobilization. Mass wasting constitutes the gravitationally driven, downslope movement of weathered regolith and rock without a primary transporting medium (fluid). Governed by Mohr-Coulomb shear strength criteria, mass movements classify across velocity and moisture gradients into creep, solifluction, slumping, debris flows, and catastrophic rockfalls, posing acute geomorphic hazards across Pakistan's Karakoram and Himalayan highway corridors.

🎯 Exam Questions Linked

• **2022 Q.1:** "Discuss various processes of physical and chemical weathering with support from sketches/diagrams. Explain their geomorphic significance. (20 Marks)"

• **2018 Q.1:** "Discuss the endogenic and exogenic geomorphic processes in detail. How do they shape the Earth's relief? (20 Marks)"

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WEATHERING VS. EROSION DISTINCTION: Weathering is an in-situ (static) breakdown of rocks in contact with the atmosphere, hydrosphere, and biosphere. Erosion is the dynamic mobilization and transportation of weathered debris by mobile agents (running water, glacial ice, wind, waves).

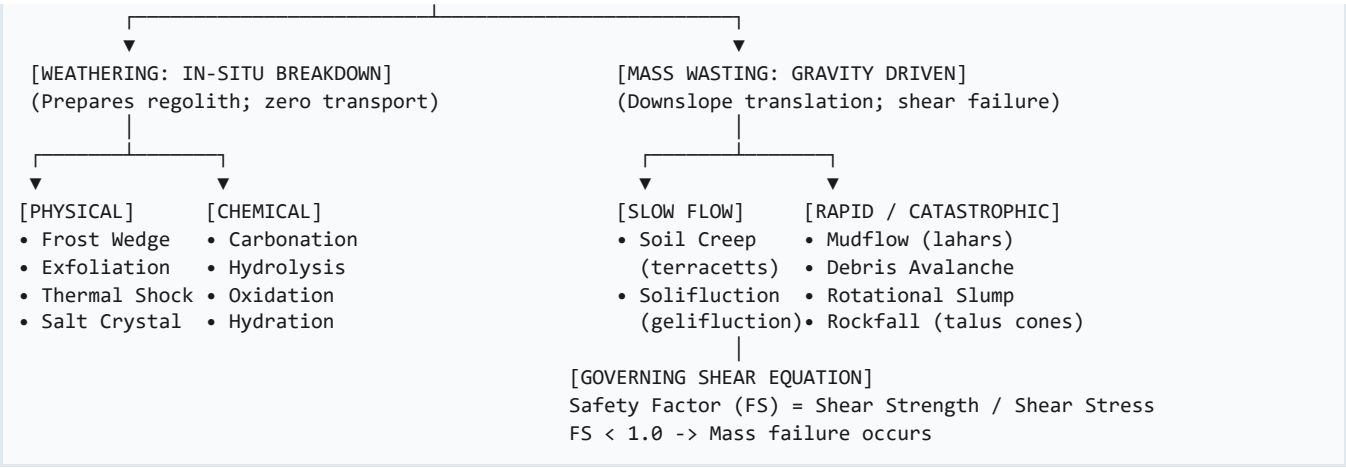
PHYSICAL WEATHERING MECHANISMS: Operates without chemical change; dominant in arid and periglacial climates: (1) Frost Wedging (9% volumetric expansion of water freezing in joints); (2) Thermal Expansion/Insolation weathering (differential mineral expansion leading to granular disintegration); (3) Exfoliation/Unloading (sheeting in granites due to overburden removal); (4) Salt Wedging (haloclasty).

CHEMICAL WEATHERING MECHANISMS: Alters mineral composition via aqueous reactions; dominant in warm, humid tropics: (1) Carbonation (atmospheric CO_2 forming weak carbonic acid H_2CO_3, dissolving limestone $CaCO_3$); (2) Hydrolysis (water reacting with feldspar to produce kaolinite clay); (3) Oxidation (ferrous Fe^{2+} rusting into ferric Fe^{3+} oxides/hematite); (4) Hydration.

MASS WASTING CLASSIFICATION (SHARPE & VARNES): Governed by water content and velocity: Slow Flowage (soil creep, solifluction in permafrost); Rapid Flowage (earthflows, mudflows, debris avalanches); Slides (translational rockslides, rotational slumps with curved shear planes); Falls (free-fall rockfalls producing talus/scree cones).

1. Conceptual Mind Map & Thematic Roadmap

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                    [EXOGENIC DENUDATION DYNAMICS]
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2. Analytical Dimensions & Geomorphic/Systemic Matrix

Geomorphic Process / Class	Physical / Chemical Mechanism	Diagnostic Landforms, Field Expressions & Pakistan Examples
Physical: Frost Wedging (Cryofracture)	Water in bedrock joints expands by 9% upon freezing, exerting up to 2,100 kg/cm^2 pressure.	Felsenmeer (block fields), angular jagged peaks, scree/talus slopes flanking high mountain roads (e.g., Karakoram Highway / Babusar Pass).
Physical: Exfoliation / Unloading	Erosion strips deep lithostatic overburden, causing plutonic rocks to expand upward along release joints.	Exfoliation domes, sheeting joints, curved rock slabs peeling like onion layers (e.g., granitic domes of Nagar and Swat Valley).
Chemical: Carbonation & Solution	CO2 dissolves in rainwater: H2O + CO2 -> H2CO3; reacts with calcite: CaCO3 + H2CO3 -> Ca(HCO3)2 (soluble).	Karst landscapes: sinkholes, solution grikes/clints, subterranean cavern systems (e.g., Margalla limestone caves, Salt Range).
Chemical: Hydrolysis & Oxidation	Hydrolysis: K-feldspar + H+ + H2O -> Kaolinite clay + K+ + Silica. Oxidation: 4FeSiO3 + O2 -> 2Fe2O3 + 4SiO2 (rusting).	Saprolite deep weathering profiles, spheroidal weathering boulders, lateritic red soils in humid subtropical climates; structural weakening of bedrock.
Mass Wasting: Rotational Slump & Rockfall	Shear stress exceeds shear strength; rotational shear failure along concave slip plane; free-fall of detached blocks.	Crescentic head scarps, backward-tilted bench blocks, toe hummocks, extensive talus cones (e.g., Attabad Landslide 2010, Hunza, blocking river).
Mass Wasting: Debris Flows & Mudflows	Water-saturated regolith loses internal friction; behaves as high-density fluid moving down channels at 50 km/h.	Alluvial debris fans, structural burial of settlements and bridges along active glaciated Himalayan/Karakoram valleys during monsoon deluges.

3. 20-Mark Master Model Essay

Introduction: The Denudational Engine of the Earth's Surface

The morphology of the continental lithosphere represents a continuous battlefield between endogenic forces, which construct relief through tectonic orogeny and volcanism, and exogenic processes, which seek to denude elevated topography back to base level. Denudation operates via a fourfold continuum: weathering, mass wasting, erosion, and deposition. Weathering stands as the foundational preparatory process—the in-situ (static) mechanical disintegration and chemical decomposition of exposed bedrock under atmospheric, hydrospheric, and biological conditions, without involving an active fluid transporting medium. Once rock cohesion is broken down into unconsolidated regolith, Mass Wasting mobilizes this material downslope under the direct pull of terrestrial gravity. Together, weathering and mass wasting govern slope evolution, soil genesis, and sediment supply for global drainage basins, while simultaneously constituting severe geomorphic hazards in high-relief orogenic belts like Pakistan's northern mountain arc.

Mechanisms of Physical (Mechanical) Weathering: Disintegration Without Mineral Alteration

Physical weathering breaks cohesive bedrock into progressively smaller, angular fragments without altering its underlying chemical or mineralogical composition. It dominates in arid, hyper-arid, alpine, and polar environments

where diurnal temperature fluctuations and freeze-thaw cycles are acute. The primary physical mechanism is 'Frost Wedging' (cryofracture). When liquid water enters micro-fractures and freezes at 0°C, it expands by approximately 9 percent in volume, exerting tensile pressures exceeding 2,100 kilograms per square centimeter—far surpassing the tensile strength of any rock. This produces jagged felsenmeer (block fields) and feeds extensive talus/scree cones along mountain flanks. In hyper-arid desert basins, 'Thermal Expansion' (insolation weathering) drives mechanical disintegration: because mineral grains (quartz, feldspar, biotite) possess unequal coefficients of thermal expansion, intense diurnal heating and nocturnal cooling generate internal stresses, producing granular disintegration and exfoliation. Furthermore, 'Pressure Release / Unloading' occurs when erosional stripping removes kilometers of overlying lithostatic weight, causing underlying plutonic granites to expand elastically and fracture along parallel sheeting joints, forming massive exfoliation domes.

Mechanisms of Chemical Weathering: Molecular Decomposition and Mineral Synthesis

In contrast, chemical weathering involves the complex molecular alteration of primary rock minerals, transforming them into secondary minerals (such as clays and iron oxides) or soluble aqueous ions. It reaches maximum intensity in warm, humid equatorial and subtropical zones where water, atmospheric oxygen, and biological carbon dioxide are abundant. The dominant chemical process is 'Carbonation', wherein atmospheric or soil CO2 dissolves in meteoric rainwater to form weak carbonic acid (H2O + CO2 -> H2CO3). This acidic groundwater attacks calcium carbonate in limestones (CaCO3 + H2CO3 -> Ca(HCO3)2), converting insoluble rock into soluble calcium bicarbonate—the geochemical foundation of worldwide Karst topography. Second, 'Hydrolysis' represents a chemical reaction where water molecules dissociate into H+ and OH- ions, directly attacking silicate minerals. In granites, hydrolysis leaches potassium ions from orthoclase feldspar, transforming rigid crystals into soft, expansive kaolinite clay while releasing free silica. Third, 'Oxidation' occurs when dissolved oxygen reacts with iron-bearing mafic minerals (pyroxenes, amphiboles, olivine), producing reddish-yellow ferric oxides (hematite Fe2O3 and limonite) in a process analogous to rusting, fatally compromising rock structural integrity.

Mass Wasting Mechanics: Slope Stability and the Shear Strength Threshold

Mass wasting (mass movement) encompasses all downslope movements of weathered bedrock and soil under the direct influence of gravity, distinct from fluvial or aeolian transport where a fluid medium acts as the carrying agent. Mechanically, slope stability is governed by the Coulomb-Terzaghi shear strength equation: tau = c + (sigma - u) * tan(phi), where tau is shear strength, c is cohesion, sigma is normal stress, u is pore water pressure, and phi is the internal angle of friction. A slope fails when the Factor of Safety (FS = Shear Strength / Shear Stress) falls below 1.0. Shear stress is increased by gravitational slope steepening (undercutting by rivers or road construction), tectonic seismic shaking, and heavy overburden. Conversely, shear strength is catastrophically degraded by water saturation: water lubricates slip planes, adds buoyant hydrostatic pore pressure that reduces effective normal stress (sigma - u), and adds immense gravitational weight to unconsolidated regolith.

Classification of Mass Movements: Creep, Slumping, Debris Flows, and Falls

Following the geomorphic classifications of C.F.S. Sharpe and D.J. Varnes, mass wasting is categorized across two physical parameters: velocity (slow to catastrophic) and moisture content (dry to saturated). 'Slow Flowage' is exemplified by Soil Creep—the imperceptible, millimeter-scale annual downslope migration of regolith caused by freeze-thaw heaving, evidenced by tilted fence posts, bent tree trunks, and soil terracettes. In periglacial landscapes, 'Solifluction' (or gelifluction) occurs when the active soil layer thaws over impermeable permafrost, sliding slowly downslope as lobed sheets. 'Rapid Flowage' includes Mudflows and Debris Flows, where saturated, fine-grained material behaves as a viscous liquid, surging down mountain channels at speeds exceeding 50 km/h, often triggered by monsoon cloudbursts or glacial lake outbursts (GLOFs). 'Slides' involve shear displacement along distinct failure planes: Rotational Slumps exhibit a backward-rotated head scarp along a concave slip surface, whereas Translational Rockslides move rapidly down planar bedding or joint surfaces. Finally, 'Falls' entail the free fall of detached rock blocks down vertical cliffs, accumulating as massive scree slopes.

Conclusion: Geomorphic Hazards and Slope Dynamics in Pakistan

In evaluative summation, weathering and mass wasting operate as coupled geomorphic drivers that continually sculpt the Earth's subaerial terrain. In mountainous territories like Khyber Pakhtunkhwa, Gilgit-Baltistan, and Azad Kashmir, these processes transcend academic geomorphology to present urgent socioeconomic and infrastructure hazards. The collision of the Indian and Eurasian plates maintains steep, unstable slopes, while monsoon rains and seismic tremors

frequently trigger catastrophic mass failures. A tragic manifestation was the January 2010 Attabad Landslide in Hunza, where a colossal rock avalanche severed the Karakoram Highway, dammed the Hunza River, submerged multiple villages, and formed a 21-kilometer-long lake. Managing these relentless exogenic forces requires sophisticated terrain mapping, geotechnical slope stabilization (rock bolting, drainage adits, gabion retaining walls), and strict land-use zoning based on geomorphic principles.

4. Scholarly Perspectives & Geomorphic Axioms

Scholar / Geomorphologist	Theoretical Contribution / Concept	Exact Quotable Academic Thesis / Axiom
C.F.S. Sharpe (1938)	Landslides and Related Phenomena	"*Mass wasting represents the universal gravitational adjustment of slopes, operating across a continuous spectrum from imperceptible soil creep to catastrophic rock avalanches.*"
Karl Terzaghi (1925)	Theoretical Soil Mechanics & Effective Stress	"*The catastrophic failure of slopes under saturated conditions is not caused by water acting as a simple lubricant, but by the development of neutral pore-water pressure which nullifies frictional resistance.*"
B.W. Sparks (1972)	Geomorphology	"*Weathering is the essential, indispensable precursor to all effective subaerial erosion; without chemical decay and mechanical shattering, rivers and glaciers would operate upon unyielding bedrock.*"
David J. Varnes (1978)	Slope Movement Types and Processes	"*Classification of slope failures must integrate the type of movement—fall, topple, slide, spread, or flow—with the nature of the material, whether solid bedrock or unconsolidated engineering soils.*"
Arthur N. Strahler (1952)	Dynamic Basis of Geomorphology	"*Slope forms and mass movement velocities reflect an exact equilibrium between the applied gravitational shear stresses and the resisting shearing strength of the rock-soil system.*"